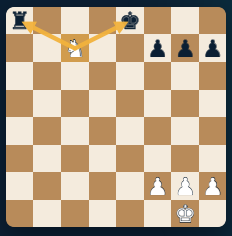
TURNING A CHESS ENGINE INTO SCIENCE AI

# Chess AI is now solving science problems.

The search that lets a computer beat chess champions now helps scientists at **Brookhaven National Laboratory** decide which measurement to take next. In one week, students build a chess engine named **Remus** in Python, then turn its thinking on a science puzzle.

5 days, 8:30 to 4:30 | All students welcome | Coding and chess

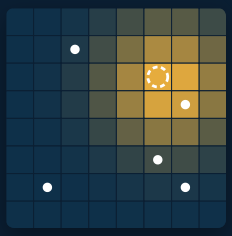


ON THE BOARD

**Which move next?**

Remus looks ahead and finds the knight fork.

SAME SEARCH: LOOK AHEAD, SCORE, CHOOSE



IN THE LAB

**Which measurement next?**

An AI picks the spot most likely to reveal the hidden peak.

## Five days, four engines EACH DAY ADDS A VERSION

DAY 1

### Remus1 plays legal chess

Legal moves at random. In Lucas Chess, you play the instructor's Remus1.

193 lines of code

DAY 2

### Remus2 searches ahead

Scoring and alpha-beta search. Remus2 plays Remus1, then you.

161 lines of code

DAY 3

### Remus3 adds the biggest gains

An opening book, a memory of positions, and the clock. Remus3 plays Remus2.

193 lines of code

DAY 4

### Remus4 adds everything else

Draws, pruning, and the endgame. Remus4 plays Remus3 and Remus4-Rust.

159 lines of code

DAY 5

### Solve a science problem

Find a hidden peak, name the crystals in an X-ray pattern, and share your results.

57 lines of code

## Every day FIVE WEEKDAYS IN A ROW

| 8:30 to 10:00 | 10:00 | 10:30 to 12:00 | 12:00 to 1:00 | 1:00 to 2:30 | 2:30 | 3:00 to 4:30 |
|---|---|---|---|---|---|---|
| Period 1 | Break | Period 2 | Lunch | Period 3 | Break | Period 4 |

Student pairs type the code and swap roles each period. Module runs check syntax and imports, usually with no output. verify_remus.py checks reference cases; success prints 'passes every test'. Lucas games use instructor engines.

YOUR SCIENCE CHALLENGE ON DAY 5

### Find the hidden peak.

Program an AI that tracks down a hidden hot spot in as few measurements as possible. Then aim your chess search at a real beamline question: which crystals are in a powder? Scientists at Brookhaven's NSLS-II ask it every day.

Take home your four engines and science programs on USB, without .venv. On a Windows computer, follow README.md setup, then the Lucas Chess guide to register and play your own engine.

## For parents

**Does a student need a laptop?**

No. Student pairs use class laptops and complete the Day 1 setup with their teacher. Twenty students work in ten student pairs, coding and playing chess together.

**Will a student need online accounts?**

No. There are no sign-ups of any kind. All work stays on the class laptops and goes home on USB.

**Does a student need to know how to code?**

Yes, the basics: functions, loops, and lists, in any language such as Java. Students also need to know how to play chess, including notation such as e4 and Nf3.

**You are ready if you**

- ✔ know some coding
- ✔ can stay focused while tracing algorithms
- ✔ know how to play chess, including notation
- ✔ want to learn how AI works